



Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, product preferences, and subscription behavior from **3,900 transactions** to guide strategic business decisions.

Dataset at a Glance

3,900

Transactions

18

Features

37

Missing Ratings

Key Feature Groups

- **Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Item, Category, Amount, Season, Size, Color
- **Behavior:** Discounts, Promo Codes, Frequency, Review Rating, Shipping

Data Preparation in Python

01

Load & Explore

Imported dataset with `pandas`; used `df.info()` and `.describe()` for initial profiling.

03

Standardize & Load

Renamed columns to `snake_case`; dropped redundant `promo_code_used`; loaded cleaned data into PostgreSQL.

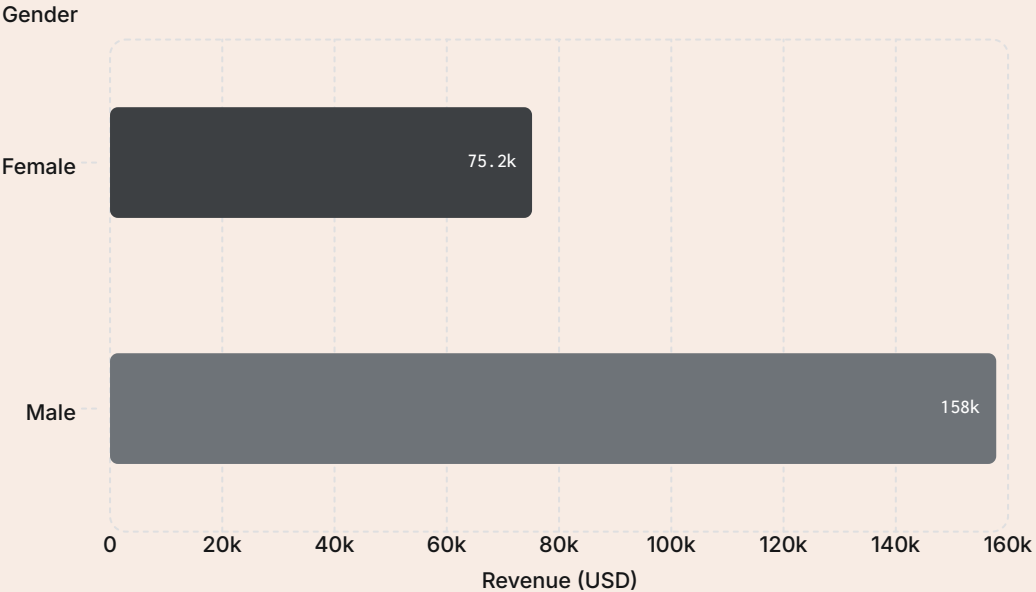
02

Clean & Engineer

Imputed missing Review Ratings with category median; created `age_group` and `purchase_frequency_days` columns.



Revenue & Spending Insights



Shipping Type vs. Spend

Shipping Type	Avg. Purchase (USD)
Standard	\$58.46
Express	\$60.48

Male customers generated **2× more revenue** than female customers. Express shipping users spend slightly more on average.

Top-Rated Products & Discount Leaders

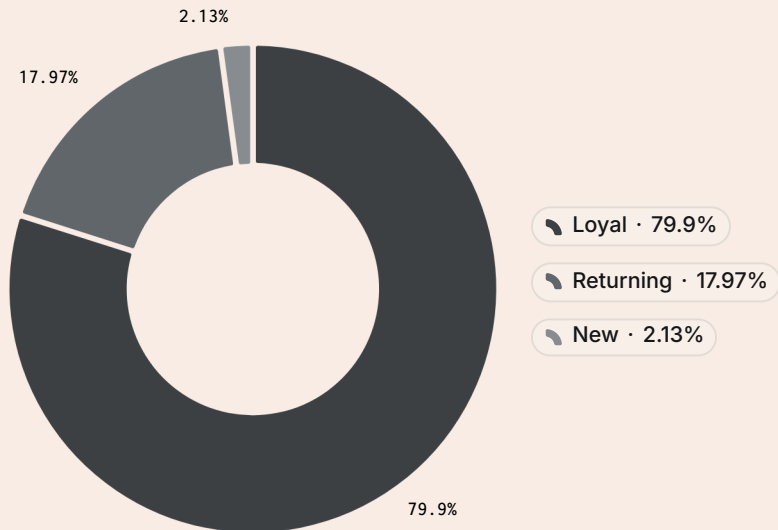
Top 5 Products by Rating

Product	Avg. Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent Products

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Customer Segmentation



Segment Breakdown

An overwhelming **80% of customers are Loyal** (>10 purchases), with 701 Returning and only 83 New customers.

Repeat Buyers & Subscriptions

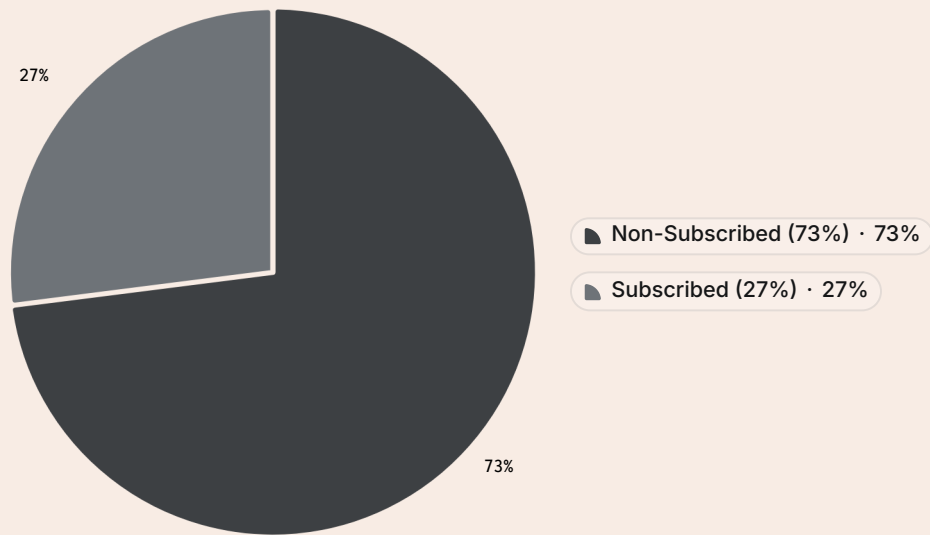
Among customers with >5 purchases: **2,518 are non-subscribers** vs. **958 subscribers** – a significant opportunity to convert repeat buyers.

Subscription Status Analysis

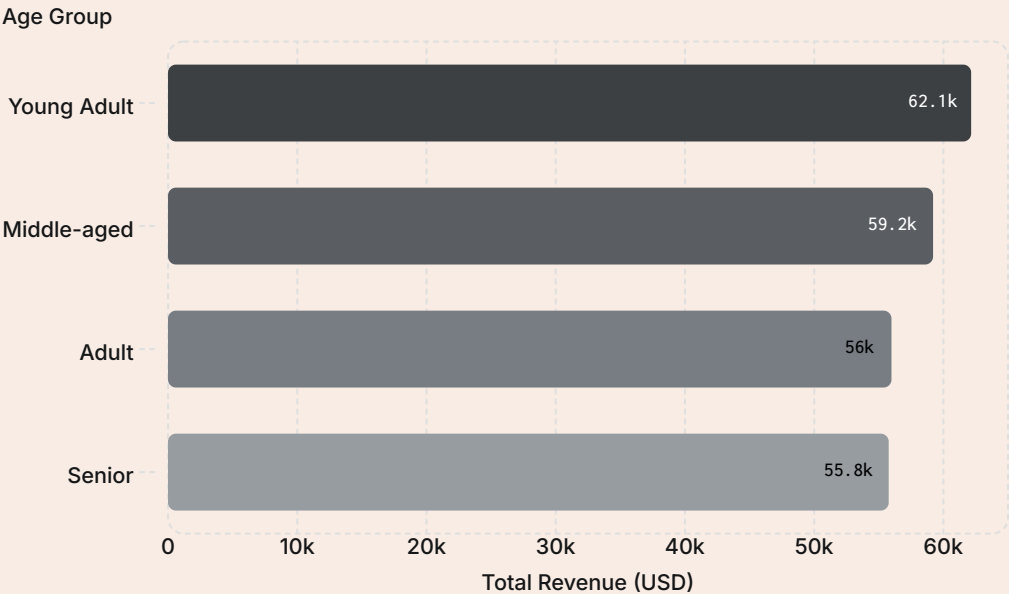
Subscribers vs. Non-Subscribers

Status	Customers	Avg. Spend	Total Revenue
Subscribed	1,053	\$59.49	\$62,645
Non-Subscribed	2,847	\$59.87	\$170,436

Average spend is nearly identical – the revenue gap is driven by **volume, not spend per visit**. Growing the subscriber base is key.



Revenue by Age Group & Top Products



Top Products per Category

Category	#1 Product	Orders
Accessories	Jewelry	171
Clothing	Blouse / Pants	171
Footwear	Sandals	160
Outerwear	Jacket	163



Power BI Dashboard

3.9K Total Customers	\$59.76 Avg. Purchase Amount	3.75 Avg. Review Rating
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Business Recommendations



Boost Subscriptions

Promote exclusive benefits to convert the 73% non-subscriber base.

Loyalty Programs

Reward repeat buyers to accelerate movement into the Loyal segment.

Review Discount Policy

Balance sales boosts with margin control – especially for Hat, Sneakers, and Coat.

Targeted Marketing

Focus on Young Adults (highest revenue) and express-shipping users for premium campaigns.